

Acoustic indices are not useful for biodiversity research

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Abstract

1. Biodiversity assessment using passive acoustic monitoring has historically been challenging due to the limited availability of multi-species acoustic detectors. In this context, acoustic indices were introduced as an alternative way to represent species diversity in acoustic datasets. Although there is increasing evidence that their effectiveness as a proxy for species diversity is limited, acoustic indices are still used and recommended for biodiversity research.
2. Acoustic indices face structural issues, with a flawed theoretical basis and weak empirical support. Additionally, limited generalizability, dependence on context and inconsistent performance make it difficult to assess their value in the inferential process. Without a clear method to map acoustic indices into meaningful biological components, their interpretability is greatly compromised.
3. Biodiversity assessment through acoustic monitoring benefits significantly from frameworks that use species as the fundamental analytical unit, even when misclassification might occur. In contrast, acoustic indices ignore species recognition due to their reliance on properties such as spectral distribution of acoustic energy and consequently limit the inference of causal links to species diversity.
4. A variety of options are available to enhance species-based biodiversity acoustic monitoring. We emphasize that manual annotation and single-species detectors have contributed significantly to the field, and effective methods for multi-species classification now exist. This has opened new paradigms and expanded opportunities for directly identifying target signals in audio data across a broad range of species.
5. In the ongoing biodiversity crisis, resources should be prioritized for tools and approaches that provide managers and stakeholders with meaningful information on the composition, structure, and function of biodiversity. Acoustic indices are not suitable as reliable biodiversity indicators, and their continued use in biodiversity research is unjustifiable. Acoustic monitoring should leverage existing and emerging methods to extract species-level data from recordings, offering more valuable and impactful contributions to ecology and conservation efforts.

KEYWORDS

acoustic monitoring, artificial intelligence, bioacoustics, diversity indices, ecological indicator, machine learning, soundscape, species diversity

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1 | OVERVIEW

Passive acoustic monitoring (PAM) has greatly advanced scientific research by allowing scientists to study sound-producing species over unprecedented spatial and temporal scales, even in remote and difficult-to-access ecosystems. PAM for biological purposes started in the 1960s, when marine researchers began using hydrophones to study cetaceans, especially large whales (Payne & McVay, 1971). Since then, sounds from most of the approximately 130 marine mammal species living in the world's oceans (Webber et al., 2016) have been documented (Sayigh, 2014). Highly stereotyped vocalizations, such as those of baleen whales, can be easily identified with traditional signal processing methods (Baumgartner & Mussoline, 2011). By measuring vocal activity, cue rates and signal localization, we can use advanced analytical techniques such as population density estimation to monitor populations comprehensively (Marques et al., 2013).

Although acoustically derived population estimates and behavioral studies have been possible for many marine mammal species, this has not been the case for most freshwater and terrestrial species and ecosystems until recently.

Advances in digital audio recording technologies, data storage and automated analysis ushered in the modern era of PAM in the 1990s and early 2000s. These developments allowed for broader applications (Pavan et al., 2022), including studies of coral reefs (Jarriel et al., 2024) and terrestrial ecosystems, such as tropical rainforests (Wrege et al., 2017). In these settings, audio recordings are typically complex and include many more biological sound sources than deep-ocean recordings. For example, coral reefs host the most diverse fish assemblages anywhere on Earth (6000–8000 species; Lieske & Myers, 2002), many of which are soniferous (Rice et al., 2022). Similarly, tropical rainforests are dominated by insects, with thousands of species across orders such as Orthoptera (crickets, katydids), Cicadomorpha (cicadas) and some Coleoptera (beetles) that probably produce sound (Stork, 2018; Symes, Kittelberger, et al., 2022; Symes, ter Hofstede, et al., 2022). However, most of these acoustic signals are still undescribed or linked to specific species, making it difficult to extract species-specific information from soundscape datasets. Even for birds, whose sounds are better documented, semi-automated detection from large terrestrial audio collections has remained technically challenging and was mostly limited to a few species until recently (eg Wood, Popescu, et al., 2019). Bioacousticians outside of cetacean research have faced a common problem: many biological sound sources with unknown identities and few tools for large-scale, multi-species studies (Sugai et al., 2019; Sugai et al., 2020).

In 2008, acoustic indices were introduced as a method to estimate species diversity from acoustic data (Sueur et al., 2008). These indices are mathematical representations of audio data, calculated based on user-selected frequency ranges and time intervals (Sueur et al., 2014). Initially, it was assumed that acoustic indices would vary predictably across space and time due to differences in community structure and composition, serving as a proxy for animal biodiversity. Unlike other biodiversity methods that rely on a fundamental biological unit (often at the species-level), acoustic indices rely on mathematical summaries

of sound energy. This shift in the core, fundamental unit complicates the assessment of key biodiversity components such as taxonomic or functional diversity. Additionally, a comprehensive meta-analysis found that the relationship between acoustic indices and species diversity metrics is weak and inconsistent (Alcocer et al., 2022). While using acoustic indices for biodiversity assessment appeared reasonable when alternatives were limited, new machine-learning-based tools now address many of their limitations and enable more robust, comprehensive, species-based assessments of animal diversity (Sethi et al., 2024; Wood et al., 2024). Recent advancements also extend the application of existing machine-learning algorithms to different taxonomic groups (Ghani et al., 2023; Kim et al., 2023) and broader soundscapes (Parcerisas et al., 2024; Sethi et al., 2020).

Despite mounting evidence that acoustic indices are insufficient as proxies for species diversity and the availability of accessible tools for species-based biodiversity assessment, acoustic indices are still recommended as a reliable method for PAM-based biodiversity monitoring (eg, Bradfer-Lawrence et al., 2023, 2025; Ross et al., 2023). Our goal with this Perspective is to highlight the limitations of acoustic indices as proxies for biodiversity and to suggest alternative approaches that can produce robust biodiversity data. Whether acoustic indices remain promising for ecological research or not, detecting and understanding changes in biodiversity is much more meaningful when based on species-specific metrics rather than on mathematical abstractions.

2 | WHY ARE ACOUSTIC INDICES LIMITED FOR BIODIVERSITY INFERENCE?

2.1 | Acoustic indices lack theoretical and empirical support as proxies for biodiversity

The core theory underlying the biological relevance of acoustic indices is the acoustic niche hypothesis (Krause, 1993), which posits that time and frequency domains are partitioned between sound-producing species. Assuming this hypothesis, one can deduce that the diversity of acoustic information is related to the diversity of vocalizing species. Yet, it only takes one spectrogram from a tropical soundscape to refute this prediction: overlapping vocalizations in frequency and time, between and within biological groups (eg birds, frogs, insects) are very common, irrespective of the diversity of species (Figure 1). Additionally, integrative approaches also showed that the acoustic output of ecological communities is inconsistent with predictions of a partitioned acoustic space (Sugai et al., 2021; Tobias et al., 2014). Therefore, a positive relationship between species diversity and the distribution of acoustic energy is a misconception, exposing a critical issue with the core theoretical framework of acoustic indices.

The theoretical inadequacy of acoustic indices as biodiversity proxies manifests in widespread poor performance: The relationship between acoustic indices and species diversity is consistently low, irrespective of index, taxa or the diversity dimension represented (Alcocer et al., 2022). Moreover, researchers are not evaluating the efficacy of indices: 75% of published articles did not test the relationship



2.2 | Acoustic indices lack generalizability

Mathematical indices have a long history in ecology, and the evolution of acoustic indices has commonalities with the trajectory of landscape indices. A myriad of landscape metrics are readily available to quantify spatial patterns (eg FRAGSTATS, APAK) and have been widely applied in studies oriented towards landscape heterogeneity (Tonetti et al., 2023). However, most of these metrics are sensitive to attributes such as spatial scale and landscape configuration and composition, which collectively limit their generalizability (Li & Wu, 2004). Similarly, scaling and generalizability issues exist with acoustic indices. Evidence from studies all around the world shows that no single index or combination of indices best correlates to species diversity, with positive relationships contingent on taxa, ecosystem and the dimension of species diversity addressed (Bicudo et al., 2023; Dröge et al., 2021; Giuliani et al., 2024; Kotian et al., 2024; Mammides et al., 2017; Sethi et al., 2023). For acoustic indices to be comparable across space and time, generalizability is a critical requirement. Because the transferability and generalizability of the same index across geographies and ecosystems are limited, a data dredging approach is rather common: a large set of indices is related to species diversity and selected based on the highest correlation achieved (Buxton et al., 2018; Dröge et al., 2021). However, ensemble approaches that combine multiple indices are subject to inflated Type 1 error rates, which ultimately affect transferability (Allen-Ankins et al., 2023; Bicudo et al., 2023; Kotian et al., 2024).

As noted above, the poor performance of acoustic indices as biodiversity proxies may arise from inappropriate implementation: among a series of practical considerations, using temporal subsampling and different resolutions inferred or assumed from 'ecological context' are suggested as best practices (Bradfer-Lawrence et al., 2023; Metcalf et al., 2021). Yet, there is no support for relying on any particular temporal resolution or subsampling approach based on the relationship between acoustic indices and species—further underscoring the lack of transferability (Bicudo et al., 2023). Collectively, the evidence points to the conclusion that acoustic indices are consistently inconsistent across ecosystems. With no generalizability and no predictive diversity power, applying acoustic indices to audio data must start from scratch at every new site, or even with each new ARU deployment, because results from other places, times or taxa cannot be readily transferred across different conditions.

Lessons from the more mature field of landscape indices also apply to bioacoustics: indices, whether representing the landscape or the soundscape, should be selected for their ability to identify ecologically relevant spatial patterns rather than being cherry-picked through p-hacking (Popovic et al., 2024; Steel et al., 2013).

2.3 | Acoustic indices are difficult to interpret (and therefore, are not actionable)

In a time of rapid global change, natural resource managers make management decisions amid great uncertainty and limited resources (Polasky et al., 2011). There are widespread and significant changes in

acoustically detectable taxa, from birds (Lees et al., 2022; Rosenberg et al., 2019) to amphibians (Luedtke et al., 2023) and insects (Wagner et al., 2021). Therefore, interpretable biodiversity monitoring methods are essential to support evidence-based decisions to conserve these taxa and the ecological processes they sustain. Decision-making processes must be transparent, replicable, defensible and easy to explain to project partners to build trust and ensure lasting success (Fuller et al., 2020). However, not all biological metrics are suitable for measuring a research target, and ambiguity in mathematical indices—as opposed to concrete population metrics—can increase conflict between partners (Elbroch et al., 2022).

Since acoustic indices are already difficult for researchers to use (Section 2.2), they can be even harder for third parties to interpret. Guidance warns against using acoustic indices as proxies for biodiversity (eg species richness) because these indices need specialized knowledge to interpret (Bradfer-Lawrence et al., 2025) and are not intuitive (Gibb et al., 2024). Relying on acoustic indices to measure biodiversity adds ambiguity to scientific and decision-making processes that are already filled with uncertainty. Continuing to use acoustic indices in conservation or management work is an avoidable but common distraction from more understandable taxonomically based acoustic detection methods.

3 | WHY IS SPECIES-BASED ACOUSTIC MONITORING BEST FOR BIODIVERSITY RESEARCH?

3.1 | Inference on biodiversity requires biological grounding

Biodiversity is a concept that includes variability across multiple biological levels, taxonomic units, functional groups, phylogenetic histories and the environments species inhabit (IPBES, 2019). It is inherently multidimensional, requiring multiple descriptors to capture the phenotypic, functional and phylogenetic attributes of various species (Magurran, 2013). However, documenting all species in a given area is nearly impossible (but see Ruppert et al., 2019), so a common practice is to focus the analyses on a specific taxonomic group, such as mammal or bird assemblages. In practice, even surveying a significant proportion of a prominent taxonomic group such as birds can be challenging, and additional resolutions might be necessary (eg diurnal bird assemblages). Aside from defining the unit of analysis for biodiversity assessment, measuring the multifaceted dimension of the diversity entity is challenging (Magurran, 2013). Regardless of the methodological framework, biodiversity metrics should be based on species or other meaningful taxonomic/biological units as the fundamental unit.

Acoustic indices focus on soundscapes and/or ecological communities, but unlike species-based biodiversity methods, the fundamental unit of analysis is not a biological entity, as the recognition of species is presumably unnecessary (Bradfer-Lawrence et al., 2025). Statistical summaries of sound pressure levels are

used to represent a wide range of species occurrences. They act as a biodiversity heuristic based on mathematical abstractions, without relying on any taxonomic units (Figure 1), and often rest on flawed foundational assumptions (see Section 2.1). The absence of species-based reference points creates a gap between acoustic indices and their biological proxies, limiting ecological interpretation. A change in the value of an acoustic index over time usually has little direct biological meaning; in contrast, changes in species richness, community composition or population size has clear biological implications (Figure 2).

The field of ecology already faces challenges with variation in effect sizes caused by differences in researchers' analytical choices (Gould et al., 2025). Adding an extra layer of complexity by selecting acoustic indices over species-based methods exacerbates this problem, as it introduces a mathematical abstraction as the fundamental measurement unit. Conversely, species-based detection methods in acoustic monitoring—whether through manual annotation, band-limited energy detection, cross-correlation, machine-learning models or other techniques (see Section 4)—offer a starting point to evaluate various biodiversity metrics such as taxonomic, phylogenetic and functional diversities (Moreno et al., 2017), or even surrogate species approaches (eg Brunk et al., 2025).

Altogether, and even if there were no alternatives to multi-species automated detection, there is sufficient evidence to disqualify acoustic indices as a reliable tool for measuring biodiversity (Sugai & Costa-Pereira, 2025).

3.2 | Challenges in species identification do not impede species-based approaches

Scaling acoustic monitoring using species-based approaches entails understanding the consequences of decisions made during different stages of analysis (discussed in Section 4, Figure 2). For example, converting a prediction into a confirmed species record using a classification threshold, or validating a detector or manually identifying species based on knowledge about their vocal repertoire, acoustic features of their vocalizations, and vocalization behavior. During this process, both humans and machines can make mistakes, which can negatively affect the validity of observed patterns. There is extensive literature covering the effects of both false negatives and false positives (Bailey et al., 2014; MacKenzie et al., 2002, 2017; Tire et al., 2003), providing valuable guidance for handling imperfect detection from detectors applied to acoustic data. False positives can be entirely prevented through manual review of detector predictions; however, this process can be time-consuming and labor-intensive. Additionally, advanced methods are available to address false-positive errors in occupancy models that use acoustic data (Chambert et al., 2018; Rhinehart et al., 2022; Wood et al., 2024). There are also modeling frameworks designed to thoughtfully address imperfect detection using species diversity, species distributions, and communities (Guillera-Aroita, 2017; Iknayan et al., 2014; Richter et al., 2021).

Human-validated species detections can be integrated into diversity modeling using defensible and understandable methods (Knight et al., 2017; Wood & Kahl, 2024). A clear set of methodological guidance for applying acoustic indices to biodiversity assessment has not yet been established, as users of acoustic indices are often advised to ground truth and validate their acoustic data. We believe such efforts are more effective when focused on producing interpretable results that directly relate to the research question of interest.

4 | HOW SPECIES-BASED ACOUSTIC MONITORING ENABLES BIODIVERSITY INFERENCE

Many approaches exist for extracting species detections from PAM, but only recently have they scaled to both high audio volumes and large numbers of species. We briefly outline these options to show how some species-based approaches to biodiversity research are now achievable with PAM data.

The most basic species-based PAM approach involves trained analysts manually annotating subsets of audio recordings to identify species of interest. This was the main analytical method in terrestrial acoustic monitoring until 2019 (Sugai et al., 2019). Manual annotation for biodiversity research is time-intensive and requires meticulous data management (Clink et al., 2024). When limited time for data review is an issue, estimating species diversity with manual annotations can be optimized using species accumulation curves and species diversity estimators (Anunciação et al., 2022; Symes, Kittelberger, et al., 2022; Symes, ter Hofstede, et al., 2022). Additionally, long-term spectral averages have long been employed to analyze large datasets in marine bioacoustics applications and are similarly useful in terrestrial acoustic monitoring (Clink et al., 2020).

However, various user-friendly open-source options exist for creating and managing annotations, such as Whombat (Martínez Balvanera et al., 2025), AviaNZ (Marsland et al., 2019), AMMonitor (Clarfeld et al., 2025) and NEAL (Gibbons et al., 2023). Many other options exist (eg see <https://rhine3.github.io/bioacoustics-software/>), in addition to feature-rich platforms such as ARBIMON (<https://arbimon.org/>), Raven Pro (K. Lisa Yang Center for Conservation Bioacoustics, Cornell University) and WildTrax (<https://wildtrax.ca/>). Emerging promising alternatives include ML-based unsupervised clustering and human-in-the-loop tools (Napier et al., 2025), and multi-staged hybrid approaches (Alexander et al., 2025). Fortunately, as we describe below, bioacoustics has advanced tremendously beyond manual annotation.

4.1 | Long-standing availability of single-species tools

Classic automated detection tools such as band-limited energy detection (Gillespie & Leaper, 1996) and spectrogram cross-correlation

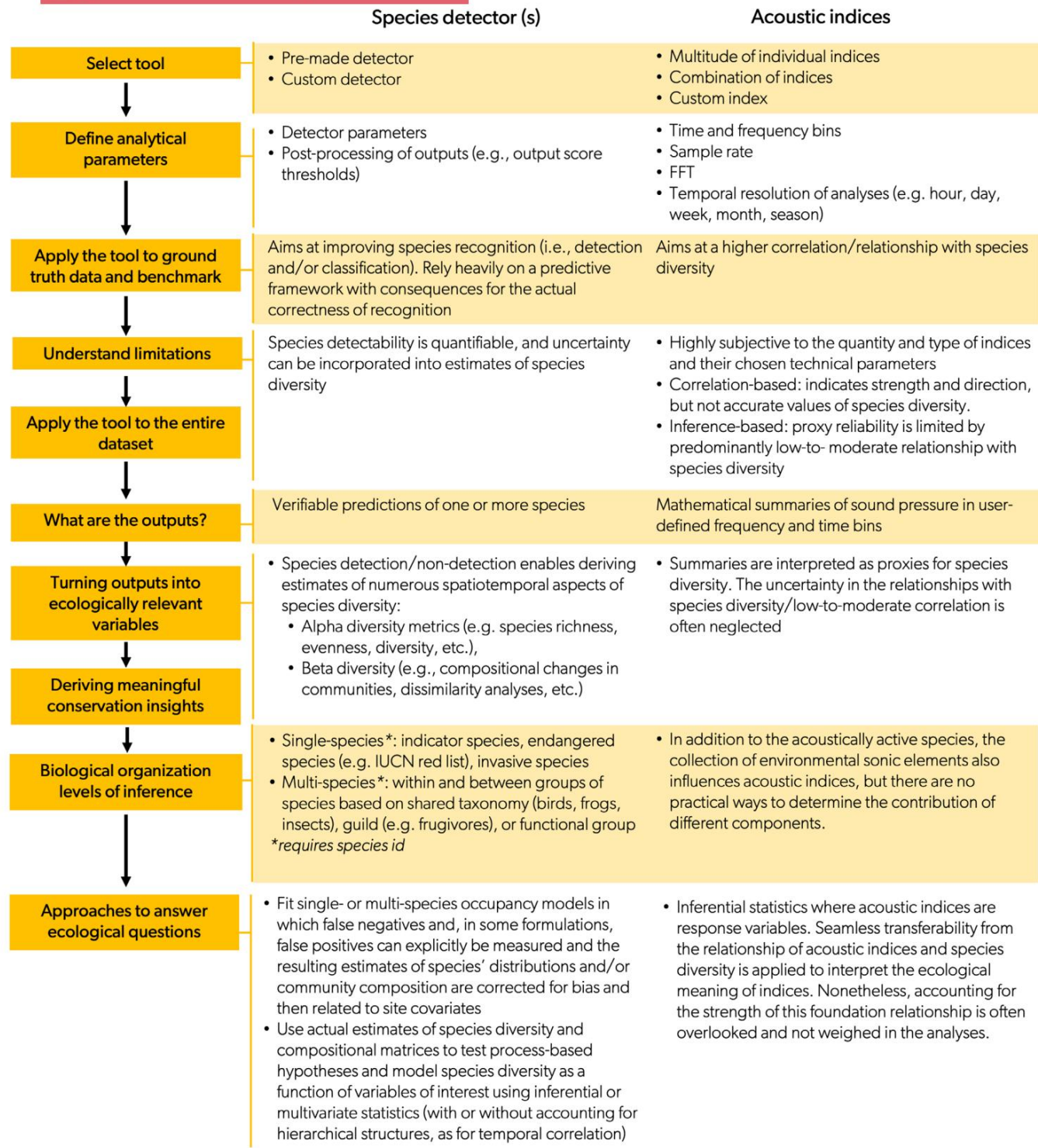


FIGURE 2 A conceptual workflow for researchers who want to assess the diversity of acoustically active taxa across some number of locations.

(Mellinger & Clark, 2000) have a well-established history of support-ing species-based automated detection in bioacoustics. Although these specialized detection methods are challenging to scale because they require a new detector for each species, they have still provided actionable insights due to their ease of biological interpretation. For example, Sebastián-González et al. (2015) used a band-limited energy

detector to assess the distribution of an endemic Hawaiian forest bird

in previously unknown locations, and Wood, Gutiérrez, et al. (2019) processed nearly 200,000 h of audio using two spectrogram-based cross-correlation-based detectors to assess population change in an invasive owl species. Overall, options for automated species detection have been available for decades, along with ecological knowledge and quantitative tools needed to make inferences about biodiversity based on the presence of specific species.

4.2 | Comprehensive multi-species tools are now available

For a long time, scalable tools for detecting animal sounds in large audio datasets were limited to only a few species. However, in recent years, deep learning has transformed wildlife monitoring through acoustic and other natively digital data (Shiu et al., 2020; Tuia et al., 2022). Analyzing large-scale acoustic datasets is rapidly shifting away from manual tagging and single-species detections to multi-species applications (Stowell, 2022; Williams et al., 2025). Machine-learning tools still have limitations, many of which relate to coverage, the quality and quantity of training (Funosas et al., 2025). Still, as ML-based tools for PAM analysis continue to advance (Sections 4.3–4.5), the potential and reach of acoustic monitoring in biodiversity research are expected to grow significantly, reaching unprecedented levels.

Convolutional neural networks (CNN) have been a successful tool for animal sound identification thanks to vast public repositories of labeled data (eg Xeno-canto, <https://xeno-canto.org/>; Macaulay Library, www.macaulaylibrary.org), with birds being the primary focal taxa. One prominent animal sound CNN is BirdNET (Kahl et al., 2021), which is freely available and the current version can predict >6500 sound classes, predominantly birds but also mammals, amphibians, and anthropogenic sounds (v2.4.0; <https://github.com/kahst/BirdNET-Analyzer>). Other high-coverage CNNs include Google Perch v2.0, trained to identify >14,000 species (van Merriënboer et al., 2025), and Nighthawk, a night flight call detector trained on 82 bird species across four orders (Van Doren et al., 2024). The collective growth of multispecies classifiers has yielded a concomitant proliferation of multispecies ecological research efforts, including, for example, 67 Use multi-species automated detection, the diversity of acoustically active species can be derived from species-based matrices.

4.3 | It is increasingly easy to create custom detectors for signals not included in pre-trained tools

Rapid innovation in machine learning has accelerated our ability to identify species with limited labeled training data, a crucial development because pre-trained multispecies classifiers tend to be biased towards birds, especially those from the Global North. New methods utilizing transfer learning leverage the knowledge from large models pre-trained on extensive datasets to recognize new classes in audio data (Pan & Yang, 2010). Early transfer learning applications involved models trained on the 'ImageNet' dataset for spectrogram classification (Dufourq et al., 2022) or pretrained models using Google's AudioSet (Çoban et al., 2020; Hershey et al., 2017). More recently, transfer learning with models trained on bird sounds (BirdNET and Perch) has achieved high performance in classifying bioacoustics events of birds, frogs, and marine mammals (Ghani et al., 2023; Williams et al., 2025).

The new task can be trained on a dataset that ranges from a few samples (Nolasco et al., 2023) to thousands (Kath et al., 2024), and applications are especially useful for rare species or species not included in pre-trained models. Data augmentation and generative AI tools can be employed to synthesize data ultimately enhancing the development of detectors for sounds with limited training data (Parry et al., 2024; Weldy et al., 2025).

The state of machine-learning technology shows rapid growth in effectively classifying acoustically active species in soundscape data. Automated classifiers from single and multiple species now provide reliable and practical ways to extract biological information from PAM recordings and close the gap between raw data and efficient natural resource management.

4.4 | Machine-learning tools are increasingly accessible

The growth of bioacoustics detection tools using deep learning, around 2016, was initially only accessible to users with significant technical skills (Stowell, 2022). The entry barrier has been greatly lowered with the release of user-friendly and free software, such as the BirdNET Analyzer graphical user interface (<https://github.com/kahst/BirdNET-Analyzer>). For example, conservation practitioners from diverse educational backgrounds have successfully used the latest transfer learning tools to create custom detectors that effectively identify rare tropical species in PAM data (eg Chi et al., 2025).

The standard pre-trained version of BirdNET, along with custom models, can also be imported into the Raven Pro software (K. Lisa Yang Center for Conservation Bioacoustics), which is a graphical user interface with a long-standing history in the bioacoustics community. ecoSound-web is an open-source, web-based graphical user interface that provides audio hosting, annotation tools and analysis with BirdNET (Darras et al., 2023). OpenSoundscape is an open-source, Python-based tool for acoustic sound detection in audio datasets, complete with detailed documentation and tutorials (Lapp et al., 2023). ARBIMON offers various automated analyses, including detection via traditional machine learning and deep learning, as well as acoustic pattern matching. The list above is not exhaustive, but it highlights the wide range of options available. Making advanced ecological tools freely accessible and user-friendly for non-experts is essential for encouraging widespread adoption.

4.5 | Next-generation tools are on the horizon

As machine-learning applications become more widespread and accessible, new challenges and opportunities are emerging to advance the field of bioacoustics. Considered the current frontier of machine learning, 'foundation models' are large deep learning models that can flexibly perform high-level tasks such as composing text (eg ChatGPT; OpenAI, 2023) and generating images (eg Stable Diffusion; Stability AI, 2023). In the domain of bioacoustics, they may offer new

possibilities for biodiversity research through their extensive sound event classification capabilities and their ability to support the development of smaller, purpose-built models (Robinson et al., 2024). van Merriënboer et al. (2024) identify model evaluation principles to guide the creation of bioacoustic foundation models, and some results suggest that BirdNET v2.4 and Google Perch already meet an important criterion: robust classification performance for taxa that were not included in the training data (Ghani et al., 2023). How might foundation models influence the future of bioacoustics?

First, performance should continue to improve for both pre-trained classes and for transfer learning-based detectors targeting new sound classes. The model distillation process, often called 'student-teacher' learning, is vital in training such application-specific models. A high-capacity, general-purpose model (the 'teacher', or foundation model) trains a more specialized, lighter model (the 'student') that focuses on a specific subset of data or specific task requirements (Gou et al., 2021; Hinton et al., 2015). In bioacoustics, this method can enable smaller, task-specific detectors by 'distilling' knowledge from a powerful foundation model, effectively transferring its learned features and processes to the student model. This approach is especially helpful for bioacoustics applications in remote or resource-limited environments, where training data may be sparse or the target class composition changes over time.

Second, foundation models may accelerate the paradigm of automated detection away from predefined sound categories toward open-ended detection of unique acoustic events that are relevant to conservation (eg species or gunshots). Currently, feature embeddings have made it possible to differentiate adult versus juvenile vocalizations (McGinn et al., 2023) and show promise for distinguishing individuals (Lakdari et al., 2024), a key goal in bioacoustics (Knight et al., 2024). The strength of foundation models will be their improved ability to fully describe input audio variations (ie with better feature embeddings), enabling sounds to be grouped based on shared acoustic properties rather than predefined labels. By emphasizing unique acoustic events rather than species-specific categories, foundation models broaden the scope of ecological inference, enabling detection of recurring sound patterns within and across known species, environmental sounds and human-made noises (eg Nieto-Mora et al., 2024).

Current species identification tools (Section 3.2) may soon be superseded by foundation models and downstream products. We suggest that hyper-diverse tropical ecosystems should be priority candidates for the application of cutting-edge bioacoustic tools, such as foundation models (Ewers, 2024), rather than a last bastion of outdated tools, such as acoustic indices, particularly because they face critical threats globally, including cryptic extinctions (Wilson, 2017).

5 | THE FUTURE OF BIOACOUSTIC-ENABLED BIODIVERSITY RESEARCH

The continued use of acoustic indices as proxies for biodiversity is indefensible.

The rise of passive acoustic monitoring technologies has created a need for effective ways to convert large amounts of data into ecologically meaningful insights that decision-making can rely on. When acoustic indices were introduced as quick tools to summarize biological information from audio recordings, much effort was made to prove their usefulness as indicators of biodiversity.

Ironically, this effort produced significant evidence highlighting the inadequacy of acoustic indices in accurately representing biological diversity. Still, recent literature continues to support the exploration of soundscapes and their biological elements using acoustic indices for both basic and applied ecology. These claims are based on shaky foundations, misinterpreted findings and are fundamentally disconnected from species-based research. The problems with how acoustic indices are conceived and used, their limited ability to generalize, and their high level of abstraction—which hampers biological understanding—raise multiple concerns about their suitability for biodiversity studies.

Because conservation actions can be undermined by unreliable data (Popovic et al., 2024; Steel et al., 2013), using acoustic indices to guide ecological inferences and decision-making in conservation should be avoided. Accurate species identification is essential for two key conservation tasks: detecting invasive species and monitoring population trends. Both have used automated detectors for decades (eg Mellinger & Clark, 2000) and are now expanding with machine-learning-based detectors (eg Kelly et al., 2023). At the 'biodiversity' level, assessing and monitoring aspects such as functional diversity is considered a priority (Cadotte et al., 2011), while beta-diversity plays an important role for regional conservation planning (Socolar et al., 2016). Again, neither approach is practical with acoustic indices because species identities are unknown. We face a global biodiversity crisis, and relying on outdated tools to inform conservation research wastes precious resources—most notably time. We strongly urge practitioners and researchers to stop using or endorsing acoustic indices as measures of biological diversity without solid evidence of their effectiveness in supporting ecological and conservation efforts.

Fortunately, the pitfalls associated with using acoustic indices are avoidable. Essentially, we recommend that biodiversity research focus on species or context-specific relevant taxonomic groups as the core, fundamental unit of analysis, rather than relying on summary statistics of sound pressure levels. The tools for signal processing, quantitative techniques, and ecological theory to support species-based research have been available for decades and are continually improving. The development of more accurate and accessible machine-learning tools now allows for more direct and comprehensive measurement of biodiversity (Stowell, 2022).

The computer science community is increasingly engaging with bioacoustics-related challenges, such as BirdClef (<https://www.kaggle.com/competitions/birdclef-2025>) and BioDCASE (<https://biodesc.github.io/>), though training and access to the necessary computing power remain challenges for researchers in species-rich and highly threatened regions across the Global South (Martínez-Medina

et al., 2025). Regardless of how sophisticated the tools used to re-trieve biological information from audio recordings become, the ex-tracted information will always require thoughtful application.

Machine learning alone is not a panacea, and AI/machine-learning tools, if used, are still only one part of an ecological re-search process in bioacoustics (Kitzes et al., 2025). Domain experts should always be involved in the bioacoustic data workflow, and understanding model performance through ground-truth data re-mains a gold standard. Natural history experts should participate in validating a representative subset of machine-learning detec-tor outputs (Navine et al., 2024; Rhinehart et al., 2022; Wood & Kahl, 2024) and in deciding whether and how to use detector out-puts in subsequent analyzes (Wood & Peery, 2022). As mentioned earlier, one advantage of directly quantifying biological events is that it leverages the many existing methods for analyzing biodiver-sity data rather than creating new methods for analyzing indirect proxies. Additionally, modeling approaches are being developed to specifically address the unique features of machine-learning-derived observations (Doser et al., 2021; Fiss et al., 2024; Rhinehart

et al., 2022; Singer et al., 2024; Wood et al., 2024). Therefore, input from naturalists, ecologists, statisticians and others will be crucial to the responsible use of current machine-learning tools and future developments.

Currently, the critical process of extracting biodiversity informa-tion from large amounts of passive acoustic recordings is supported by tools that quickly and directly identify vocal species and key abiotic sounds through expert annotation, targeted detectors and pre-trained or customizable machine-learning models. The era of ab-stract biodiversity proxies in bioacoustics is over. We encourage re-searchers using PAM to focus on identifying biological events, such as species-specific vocalizations, and translating them into meaning-ful, actionable summaries to support sound ecological research and management decisions.

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The authors have no conflict of interest to declare.

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